

Worksheet 3B: Factoring and Solving Quadratics

Annotated Revision Guide

AIML Foundations Mathematics

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Part A: Common Factor and Simple Factoring (Problems 1–8)

Concept. Factoring is expansion in reverse: given a polynomial, find its multiplicative building blocks. Always check first for a **highest common factor** (HCF) that can be pulled out. Then look for the remaining structure: difference of squares, perfect square, simple trinomial.

Method. Step 1: Extract the HCF of all terms. Step 2: Inspect the remaining polynomial for a recognisable pattern. Step 3: For trinomials $x^2 + bx + c$, find two numbers with product c and sum b .

Watch out. Stopping after extracting one common factor when the remaining polynomial still factors. Always ask: “can I factor this further?” until each factor is irreducible.

Where this leads. Fully factored form of a polynomial reveals its **roots** (the values of x that make it zero). Roots are exactly the x -intercepts of the graph. This connection drives all of Worksheet 3D (graphing).

Explore further. Over the complex numbers, every polynomial of degree n has exactly n roots (counting multiplicity) — this is the **Fundamental Theorem of Algebra**, proved by Gauss in 1799. Over the reals, some roots may not exist (e.g. $x^2 + 1$ has no real roots).

1. $x^2 + 7x + 10$

Find factors of 10 summing to 7. $2 \times 5 = 10$, $2 + 5 = 7$.

Factor. $(x + 2)(x + 5)$.

Note: Check by expanding: $(x + 2)(x + 5) = x^2 + 5x + 2x + 10 = x^2 + 7x + 10$. ✓

2. $x^2 - 5x + 6$

Find factors of 6 summing to -5 . $(-2) \times (-3) = 6$, $(-2) + (-3) = -5$.

Factor. $(x - 2)(x - 3)$.

3. $x^2 + 2x - 15$

Product -15 , sum $+2$. $+5$ and -3 : $5 \times (-3) = -15$, $5 + (-3) = 2$.

Factor. $(x + 5)(x - 3)$.

Note: When the constant is negative, the two numbers have opposite signs.

4. $x^2 - x - 12$

Product -12 , **sum** -1 . -4 and $+3$: $(-4)(3) = -12$, $(-4) + 3 = -1$.
Factor. $(x - 4)(x + 3)$.

5. $2x^2 + 7x + 3$

AC method. $ac = 2 \times 3 = 6$; find product 6, sum 7: 1 and 6.

Rewrite. $2x^2 + x + 6x + 3$.

Group. $x(2x + 1) + 3(2x + 1) = (x + 3)(2x + 1)$.

Note: The AC method works for any $ax^2 + bx + c$.

6. $3x^2 - 10x + 3$

AC method. $ac = 9$; product 9, sum -10 : -1 and -9 .

Rewrite and group. $3x^2 - x - 9x + 3 = x(3x - 1) - 3(3x - 1) = (x - 3)(3x - 1)$.

7. $x^2 - 16$

Recognise. Difference of squares: $x^2 - 4^2$.

Factor. $(x - 4)(x + 4)$.

Note: Always check for difference of squares before trying other methods.

8. $4x^2 - 25$

Recognise. $(2x)^2 - (5)^2$.

Factor. $(2x - 5)(2x + 5)$.

Part B: Solving Quadratic Equations (Problems 9–18)

Concept. A quadratic equation $ax^2 + bx + c = 0$ has at most two solutions. Methods: (1) **Factor and use zero product property:** if $AB = 0$ then $A = 0$

or $B = 0$. (2) **Quadratic formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. (3) **Complete the square:** convert to $(x + p)^2 = q$ form.

Method. Prefer factoring when it is obvious (integers). Use the quadratic formula when coefficients are awkward or roots are irrational. The **discriminant** $\Delta = b^2 - 4ac$ tells you in advance: $\Delta > 0$: two real roots; $\Delta = 0$: one repeated root; $\Delta < 0$: no real roots.

Watch out. You can only use zero product property when one side is **zero**. $x(x + 1) = 6$ does *not* mean $x = 6$ or $x + 1 = 6$. Move everything to one side first: $x^2 + x - 6 = 0$, then factor $(x + 3)(x - 2) = 0$.

Where this leads. The quadratic formula is derived by completing the square — it is not magic, just algebra. Completing the square also puts parabolas into vertex form $(x - h)^2 + k$, which is the key to Worksheet 3D's graphing problems.

Explore further. The discriminant $b^2 - 4ac$ is related to the *discriminant* of algebraic number fields, a deep object in number theory. For more: Stewart & Tall, *Algebraic Number Theory*, Ch. 1. For now, the geometric meaning is clear: it measures how far the parabola's vertex is from the x -axis.

9. $x^2 + 5x + 6 = 0$

Factor. $(x + 2)(x + 3) = 0$.

Solve. $x = -2$ or $x = -3$.

10. $x^2 - 4x - 12 = 0$

Factor. $(x - 6)(x + 2) = 0$.

Solve. $x = 6$ or $x = -2$.

11. $2x^2 - 7x + 3 = 0$

Factor. $(2x - 1)(x - 3) = 0$.

Solve. $x = \frac{1}{2}$ or $x = 3$.

12. $x^2 = 9$

Rearrange. $x^2 - 9 = 0$, so $(x - 3)(x + 3) = 0$.

Solve. $x = \pm 3$.

Note: Or directly: $x = \pm\sqrt{9} = \pm 3$.

13. $x^2 + 6x + 9 = 0$

Recognise. Perfect square: $(x + 3)^2 = 0$.

Solve. $x = -3$ (repeated root, $\Delta = 0$).

Note: One solution with multiplicity 2 — the parabola just touches the x -axis.

14. $x^2 - 3x - 10 = 0$ using the quadratic formula

Identify. $a = 1$, $b = -3$, $c = -10$.

Discriminant. $\Delta = 9 + 40 = 49$.

Formula. $x = \frac{3 \pm 7}{2}$: $x = 5$ or $x = -2$.

Note: Check: $(5)(-2) = -10$ ✓, $5 + (-2) = 3$ ✓ — sum and product of roots.

15. $2x^2 + 5x - 3 = 0$

Formula. $a = 2, b = 5, c = -3; \Delta = 25 + 24 = 49.$

Solve. $x = \frac{-5 \pm 7}{4}$: $x = \frac{1}{2}$ or $x = -3.$

16. $x^2 - 2x - 1 = 0$

Discriminant. $\Delta = 4 + 4 = 8.$

Formula. $x = \frac{2 \pm \sqrt{8}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}.$

Note: Irrational roots: $x \approx 2.414$ or $x \approx -0.414.$

17. Complete the square: $x^2 + 6x - 7 = 0$

Move constant. $x^2 + 6x = 7.$

Complete. $(x + 3)^2 - 9 = 7$, so $(x + 3)^2 = 16.$

Solve. $x + 3 = \pm 4$: $x = 1$ or $x = -7.$

Note: Completing the square gives the vertex of the parabola directly: vertex at $(-3, -16).$

18. $x^2 + 4x + 8 = 0$

Discriminant. $\Delta = 16 - 32 = -16 < 0.$

Conclusion. No real solutions. The parabola does not cross the x -axis.

Note: Complex roots: $x = -2 \pm 2i.$ These exist in the complex number system.